

Smart Mobile Keyboard Prediction System

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RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/
K.RAMESH/OneDrive/Documents/booking/q1.py

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1. Bigram MLE
P(science | data) = 1.0

2. Backoff Model
Sequence: data science improves
Backoff level: Unseen
Probability: 0

3. Deleted Interpolation
Trigram probability = 0.667
Bigram probability = 0.667
Unigram probability = 0.167
Interpolated probability = 0.567

4. Entropy
P(is) = 0.667
P(drives) = 0.333
Entropy = 0.918 bits

5. Next Word Prediction
After: science
is -> 0.667
drives -> 0.333
Predicted next word: is

PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
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AI-Powered Customer Support Chatbot

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PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/
K.RAMESH/OneDrive/Documents/booking/q2.py

1. POS TAGGING

Sentence 1:
Book -> VB
a -> DT
flight -> NN
ticket -> NN
now -> RB

Sentence 2:
This -> DT
book -> NN
is -> VBZ
interesting -> JJ

2. HMM PROBABILITY
P(Start -> VB) = 0.5
P(book | VB) = 0.6
P(Start -> NN) = 0.5
P(book | NN) = 0.4

Probability of book as VB: 0.3
Probability of book as NN: 0.2
HMM Prediction: VB
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News Analytics and POS Tag Correction System

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PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/
K.RAMESH/OneDrive/Documents/booking/q3.py

Total frequency = 1160
economic : Count = 120 , Probability = 0.1034 , Percentage = 10.34 %
growth : Count = 450 , Probability = 0.3879 , Percentage = 38.79 %
increases : Count = 210 , Probability = 0.181 , Percentage = 18.1 %
employment : Count = 380 , Probability = 0.3276 , Percentage = 32.76 %

Most frequent word: growth
Frequency: 450

5. ENTROPY BEFORE CORRECTION
P(NNS) = 0.5
P(VBZ) = 0.5
Entropy = 1.0 bits

6. ENTROPY AFTER CORRECTION
P(NNS) = 0
P(VBZ) = 1
Entropy = 0 bits

7. TAGGING ACCURACY
Accuracy before correction = 75.0 %
Accuracy after correction = 100.0 %

8. FINAL ANALYSIS
Initial:
economic/JJ growth/NN increases/NNS employment/NN
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